Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering
24 Series

Course No.: CSE 1203
Course Title: Object Oriented Programmming

Experiment No. : 01
Experiment Title : Class & Objects

Submitted by:	**Submitted to:**
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Date of Experiment :
Date of Submission : 26 April 2026

Module 1: [Based on class & objects]

Problem Statement: Write a C++ program using class and objects to automate the banking process with following constraints:

i. A new account can be created and an old account can be deleted
ii. An account holder can debit/credit money
iii. An account can't be deleted if money exists in the account
iv. The program should be following menu operated
v. The private data members of an account holder's class are "Account No", "Name", "Age" and "Balance".
vi. Design member methods for individual operation
vii. Produce appropriate message when required

a.The template of the class is

```
class Account{
//private data members
//public methods
};
```

b.The menu is :

```
********** Main Menu ********

   1. Open New Account
   2. Close Old Account
   3. Deposit Money
   4. Withdraw Money
   5. Check Balance
   6. Exit
      Enter Your option:(1-6):_
```

c. Few Snapshots:

1. Deposit Money

```
Enter your option(1-6):3 <enter>
Enter Account Number:12678 <enter>
Enter Amount: 5000 <enter>
Successfully deposited
Press any key go to main menu.....
```

Code:

```
#include <iostream>
using namespace std;
class Account
{
public:
    string name;
    int acn;
    int balance;

    Account()
    {
    }

    void Open_account(string n, int a, int b)
    {
        name = n;
        acn = a;
        balance = b;
    }
    void Delete()
    {
        if (balance > 0)
            cout << "Account can not be deleted. Withdraw your money first ;)" << endl;
        else
        {
            name = " ";
            acn = 0;
            balance = 0;
            cout << "Account deleted successfully" << endl;
        }
    }


    void deposit(int b)
    {
        int old = balance;
        balance = b;
        old += balance;
        balance = old;
    }


    void withdraw(int b)
    {

        int old = balance;
        balance = b;
```

```
        if (b > old)
            cout << "Insufficient balance " << endl;
        else
        {
            old -= b;
            balance = old;
        }
    }
    void Getdata()
    {
        cout << "Account number : " << acn << endl;
        cout << "Name           : " << name << endl;
        cout << "balance        : " << balance << " Taka " << endl;
        cout << ".....................\n";
    }
};
void print()
{
    cout << "************* Main Menu ************ \n";
    cout << "1. Open New Account " << endl;
    cout << "2. Close Old Account" << endl;
    cout << "3. Deposit money" << endl;
    cout << "4. Withdraw money" << endl;
    cout << "5. Check balance" << endl;
    cout << "6. Exit" << endl;
    cout << "   Enter your option (1-6): " << endl;
}
int main()
{
    int choice;
    int index = 0;
    string Name;
    int Acn;
    int Bal;
    int temp_Acn, temp_balance;

    Account b[10000];

    while (1)
    {
        print();
        cin >> choice;

        switch (choice)
        {
        case 1:
            cout << "Enter your Name: ";
            cin >> Name;
```

```
        cout << "Enter your balance: ";
        cin >> Bal;
        Acn = 1000 + index + 1;

        b[Acn].Open_account(Name, Acn, Bal);
        b[Acn].Getdata();
        index++;
        break;
    case 3:
        cout << "Enter your account number: ";
        cin >> temp_Acn;
        if (temp_Acn < (index - 1) && temp_Acn > 1000)
        {
            cout << "Enter your depost balance: ";
            cin >> temp_balance;
            b[temp_Acn].deposit(temp_balance);
            b[temp_Acn].Getdata();
            cout << "Success" << endl;
        }
        else
            cout << "Invalid account" << endl;

        break;
    case 4:
        cout << "Enter your account number: ";
        cin >> temp_Acn;
        if (temp_Acn < (index - 1) && temp_Acn > 1000)
        {
            cout << "Enter your withdrwal balance: ";
            cin >> temp_balance;
            b[temp_Acn].withdraw(temp_balance);
            b[temp_Acn].Getdata();
            cout << "Sucess" << endl;
        }
        else
            cout << "Invalid account\n";

        break;
    case 2:
        cout << "Enter your account number: ";
        cin >> temp_Acn;
        if (temp_Acn < (index - 1) && temp_Acn > 1000)
        {
            b[temp_Acn].Delete();
        }
        else
            cout << "Invalid account" << endl;

        break;
```

```
        case 5:
            cout << "Enter your account number: ";
            cin >> temp_Acn;
            if (temp_Acn < (index - 1) && temp_Acn > 1000)
            {
                cout << "Your current balance: " << b[temp_Acn].balance << " Taka" <<
endl;
            }
            else
                cout << "Invalid account" << endl;

            break;
        case 6:
            cout << "Thank you" << endl;
            return 0;
            break;

        default:
            break;
        }
    }
}
```

Output Snippet

```
1. Open New Account
2. Close Old Account
3. Deposit money
4. Withdraw money
5. Check balance
6. Exit
   Enter your option (1-6):
```

```
1
Enter your Name: Ragib
Enter your balance: 5000
Account number : 1001
Name           : Ragib
balance        : 5000 Taka
.......................
```

```
Enter your option (1-6): 3
Enter your account number: 1001
Enter your deposit balance: 6000
Account number : 1001
Name           : Ragib
Balance        : 11000 Taka
.....................
Success
```

```
Enter your option (1-6): 4
Enter your account number: 1001
Enter your withdrawal balance: 3000
Account number : 1001
Name           : Ragib
Balance        : 8000 Taka
.....................
```

```
Enter your option (1-6): 2
Enter your account number: 1001
Account can not be deleted. Withdraw your money first ;)
```

```
Enter your option (1-6): 5
Enter your account number: 1001
Your current balance: 8000 Taka
```

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Course No.: CSE 1203
Course Title: Object Oriented Programmming

Experiment No. : 02
Experiment Title : Constructor, const & static

Submitted by:	**Submitted to:**
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Date of Experiment :
Date of Submission : 26 April 2026

Module 2 [Constructor, const & static]

(for Week 2, Based on CO1)

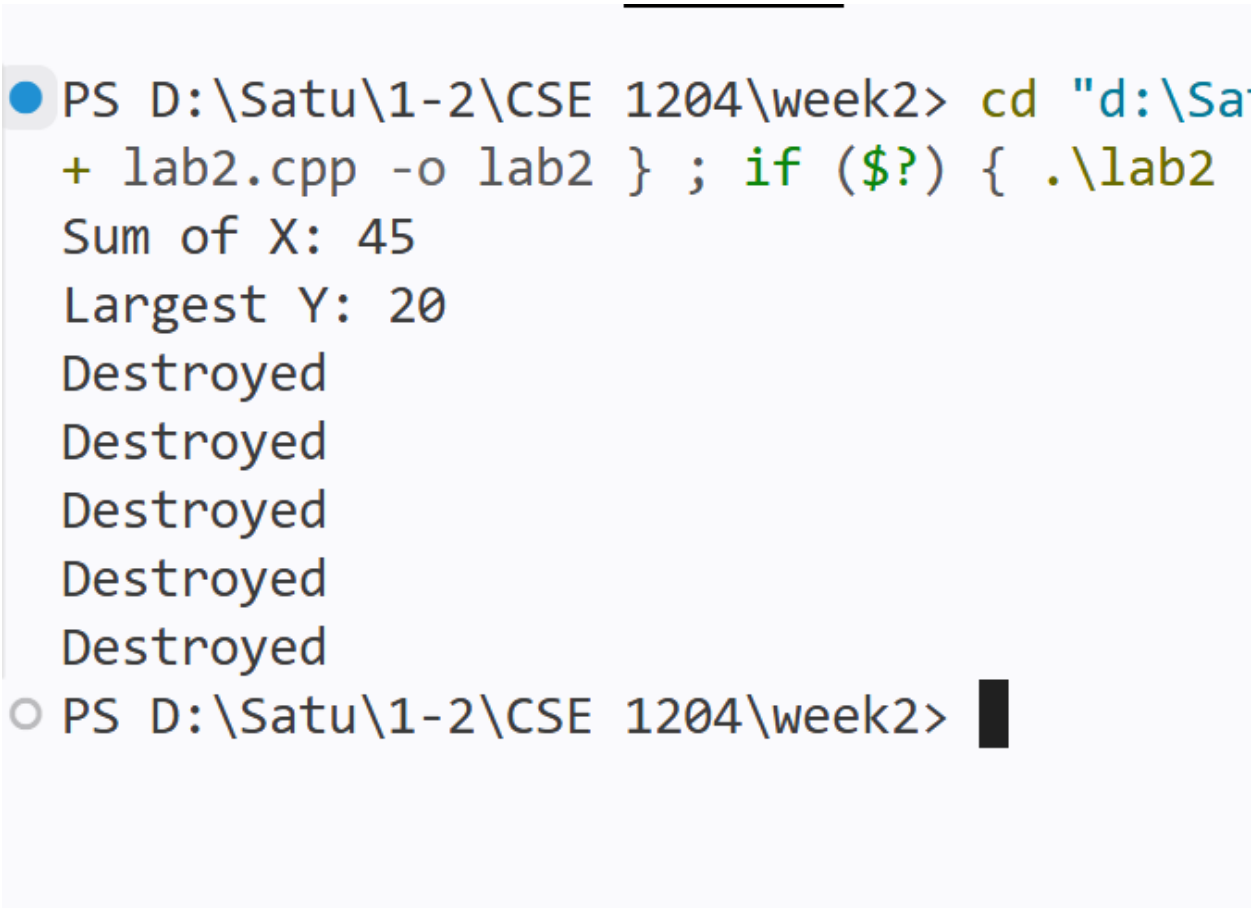
1. ***Problem Statement***: Use the following class Test with the data members and methods

```
class Test{
 private:
  Data Member x;
 Data  Member  y;
 Data  Member  z;
 public:
  //write methods

};
```

Code:

```
#include<iostream>
using namespace std;
class Test
{
    private:
            int x;
            int y;
            static int z;

    public:
            Test()
            {
                x=0;
                y=0;
                z++;
            }

            Test(int a, int b)
            {
                x=a;
                y=b;
                z++;
            }

            Test(const Test &obj)
            {
                x = obj.x;
                y = obj.y;
                z++;
            }
            void setxy(int a, int b)
            {
```

```
            x=a;
            y=b;
        }
        void display() const
        {
            cout << x << y << z << endl;
        }
        int getx()
        {
            return x;
        }
        int gety()
        {
            return y;
        }
        ~Test()
        {
            cout << "Destroyed" << endl;
        }

};
int Test::z=0;
int main()
{
    int y[10];
    Test t1;
    Test t2(5,4);
    Test t3(t2);
    Test t4,t5;

    t4.setxy(10,20);
    t5.setxy(25,14);
    int sumx = t1.getx()+t2.getx()+t3.getx()+t4.getx()+t5.getx();
    y[0]=t1.gety();
    y[1]=t2.gety();
    y[2]=t3.gety();
    y[3]=t4.gety();
    y[4]=t5.gety();
    int max = y[0];
    for(int i=0; i<5; i++)
    {
        max=(max>y[i]?max:y[i]);
    }
    cout << "Sum of X: " <<  sumx << endl;
    cout << "Largest Y: "<< max << endl;
    return 0;
}
```



Now write and necessary codes in class and **main()** do the following

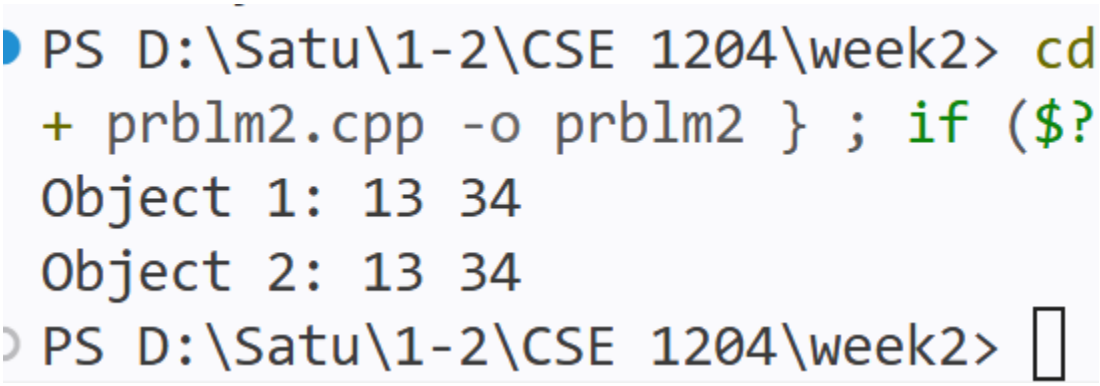
i) Initialize private data members **x** and **y** to 0 when empty constructor is called
ii) Initialize private data members **x** and **y** using parameterized constructor is called
iii) Initialize private data members **x** and y from another object using copy constructor
iv) The data member **z** keeps track of total objects created
v) Write a method to initialize **x** and **y**
vi) Write a method to display data member **z** only
vii) Write a method to display **x**,**y** and **z** where their values can't be changed
viii) Create five objects
ix) Find the sum of **x**
x) Find the object number whose **y** value is maximum
xi) Write destructor

2. ***Problem Statement*** : Write a class A with two data member **a** and **b**. Now set the values of **a** and **b** of an object **obj1** by a setter method **SetData()** and copy the values of **a** and **b** of **obj1** to another object **obj2** by using copy constructor.

```
#include<iostream>
using namespace std;
class A
{
    int a;
    int b;
```

```
    public:
            A()
            {

            }
            void setdata(int x, int y)
            {
                a=x;
                b=y;
            }
            A(const A &obj)
            {
                a=obj.a;
                b=obj.b;
            }
            void display()
            {
                cout << a << " " << b << endl;
            }
};
int main()
{
    A obj1;
    obj1.setdata(13,34);
    A obj2(obj1);
    cout << "Object 1: " ; obj1.display();
    cout << "Object 2: " ; obj2.display();

    return 0;
}
```

Output:



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Course No.: CSE 1203
Course Title: Object Oriented Programmming

Experiment No. : 03
Experiment Title : Inheritance

Submitted by:	**Submitted to:**
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Date of Experiment :
Date of Submission : 26 April 2026

Module 3 [Inheritance]:

(for Week 4 [29-3/11-12/2025)

Problem Statement: You have the create an inheritance among **`Father`**-->**Son** -->**GrandSon** class. The **`father`** class has the following data members

```
class Father{
  private:
    int money;
  protected:
    int gold;
  public:
    int land;
};
```

Now write the **Son** and **GrandSon** classes with **`private`**/**`protected`**/**`public`** access modifier and do the following:

i) Try to access **`money`**, **`gold`** and **`land`** from Son class

ii) Try to access **`money`**, **`gold`** and **`land`** from GrandSon class

iii) Find the values of money, gold and land when different access modifier is used in the following table

iv) Display the sum of money, gold and land in `Grandson` class.

Class		In Son class			In GrandSon class		
Son	**GrandSon**	**money**	**gold**	**land**	**money**	**gold**	**land**
public	public	?	?	?	?	?	?
protected	public	?	?	?	?	?	?
private	public	?	?	?	?	?	?
public	protected	?	?	?	?	?	?
protected	protected	?	?	?	?	?	?
private	protected	?	?	?	?	?	?
public	private	?	?	?	?	?	?
protected	private	?	?	?	?	?	?
private	private	?	?	?	?	?	?

Code:

```
#include<bits/stdc++.h>
using namespace std;
class Father
{   private: int money=100;
    protected: int gold=10;
    public:  int land=500;
           /*/ Father()
            {
                money=100;
                gold=10;
                land=500;
                cout << "created" << endl;
            }
            int getmoney()
            {
                return money;
            }*/
};
class Son : public Father
{
    public:
        Son()
        {

        }
        void display()
        {
        //cout << getmoney() << endl;
        cout << gold << endl;
        cout << land << endl;
        }
};
class Grandson : public Son
{
    public:
        Grandson()
        {}
        void display()
        {
            //cout << getmoney() << endl;
            cout << gold << endl;
            cout << land << endl;
        }
        //int return sum
};
```

```
int main()
{
    Son s;
    s.display();

    Grandson g;
    g.display();

    return 0;
}
```

Topic 2 [**Types of Inheritance**]: Learn and Test different types of inheritance in C++. In each inheritance draw the class diagram with class chain and try to access the data members of bases classes from child classes.

i) Single inheritance

class A{
private:
int x;
protected:
int y;
public:
int z;
}
	class B:public A{
	//write public method to //access x,y & z
	}
		int main(){
		B b;
		//call methods of class B
		return 0;
		}

Code:

```
#include <iostream>
using namespace std;

class A {
private:
    int x;
protected:
    int y;
public:
    int z;
    A()
    {

    }
    A(int a, int b, int c)
  {
    x=a;
    y=b;
    z=c;
```

```
  }
};

class B : public A {
public:
    void displayFromA(class A &obj) {
        cout << "Accessing from class B:" << endl;
        cout << "y (protected): " << obj.y << endl;
        cout << "z (public): " << obj.z << endl;
    }
    B()
    {

    }
};

int main() {
    A obj1(10,20,30);
    B obj2;
    obj2.displayFromA(obj1);
    return 0;
}
```

Multi-level inheritance

class A{
private:
int x;
protected:
int y;
public:
int z;
}
	class B:public A{
	}
		class C:public B{
		//write public //method to //access x,y & z
		}
			int main(){
			C c;
			//call //methods of //class C
			return 0
			}

Code:

```
#include<bits/stdc++.h>
using namespace std;
class A{
private:
int x=10;
protected:
int y=11;
public:
int z=12;
```

```
int getx()
{
    return x;
}
};
class B:public
A{
};
class C:private
B{
public:
int gety()
{
    return y;
}
int getz()
{
    return z;
}
int getpvt()
{
    getx();
}
};
int main(){
C c;
cout << c.getpvt() << endl;
cout << c.gety() << endl;
cout << c.getz()<< endl;
return 0;
}
```

Multiple inheritance

class A{
private:
int x;
protected:
int y;
public:
int z;
}
	class B{
	private:
	int p;
	protected:
	int q;
	public:
	int r;
	}
		class C:public A, Public B{
		//write public method //to access
		//x,y,z,p,q & r
		}
			int main(){
			C c;
			//call
			//methods of
			//class C
			return 0
			}

Code:

```
#include<bits/stdc++.h>
using namespace std;
class A{
private:
int x=1;
protected:
int y=2;
public:
int z=3;
int getx()
{
    return x;
}
};
class B{
private:
int p=11;
protected:
int q=22;
public:
int r=33;
int getp()
{
    return p;
}
};
class C:public A,public B
{
    public:
        int gety()
        {
            return y;
        }
        int getq()
        {
            return q;
        }


};
int main(){
C c;
cout << " x " <<c.getx() << endl;
cout << " y " <<c.gety() << endl;
cout << " z " << c.z<< endl;
```

```
cout << " p " << c.getp()<< endl;
cout << " q " << c.getq() << endl;
cout << " r " << c.r<< endl;
return 0;
}
```

Heirarchical inheritance

class A{
private:
int x;
protected:
int y;
public:
int z;
}
	class B:public A
	{
	//write public method to access x,y & z
	}
		class C:public A
		{
		//write method public to access x,y & z
		}
			int main(){
			B b;
			C c;
			//call //methods of //class B & C
			return 0
			}

Code:

```
#include <bits/stdc++.h>
using namespace std;

class A {
private:
    int x;
protected:
    int y;
public:
    int z;

    A() {
        x = 7;
        y = 8;
        z = 9;
    }

    int getX() {
        return x;
    }
};

class B : public A {
public:
```

```
    void showB() {
        cout << "B: x = " << getX() << ", y = " << y << ", z = " << z << endl;
    }
};

class C : public A {
public:
    void showC() {
        cout << "C: x = " << getX() << ", y = " << y << ", z = " << z << endl;
    }
};

int main() {
    B b;
    C c;
    b.showB();
    c.showC();
    return 0;
}
```

Hybrid (Diamond) inheritance [virtual class]

class A{
private:
int x;
protected:
int y;
public:
int z;
}
	class B:public A {
	
	}
		class C:public A {
		
		}
			class D:public B, public C {
			//write public method to access x,y & z
			}
				int main(){
				D d;
				//call //methods of //class D
				return 0
				}

Code:

```
#include <bits/stdc++.h>
using namespace std;

class A {
private:
    int x;
protected:
    int y;
public:
    int z;

    A() {
        x = 11;
        y = 22;
```

```
        z = 33;
    }

    int getX() {
        return x;
    }
};

class B : virtual public A {
};

class C : virtual public A {
};

class D : public B, public C {
public:
    void show() {
        cout << "x = " << getX() << endl;
        cout << "y = " << y << endl;
        cout << "z = " << z << endl;
    }
};

int main() {
    D d;
    d.show();
    return 0;
}
```

Topic 3 [Constructor & Destructor in inheritance]: Write the constructors & destructors for different types of inheritance are given as follows. Also follow and write the sequence of their execution.

i) Single inheritance

class A{
private:
int ax;
public:
//write
constructor to
initialize ax
//Write
destructor
}
	class B:public A{
	private:
	int bx;
	public:
	//write constructor to
	initialize bx
	//Write method to sum ax
	and bx
	//Write destructor
	}
		int main(){
		B b;
		//call methods of
		class B
		return 0;
		}

```cpp
#include <bits/stdc++.h>
using namespace std;

class A {
protected:
    int ax;
public:
    A() {
        ax = 10;
        cout << "A constructor\n";
    }
    ~A() {
        cout << "A destructor\n";
    }
};

class B : public A {
private:
    int bx;
public:
    B() {
        bx = 20;
        cout << "B constructor\n";
    }
    void sum() {
        cout << "Sum = " << ax + bx << endl;
    }
    ~B() {
        cout << "B destructor\n";
    }
};

int main() {
    B b;
    b.sum();
    return 0;
}
```

ii) Multi-level inheritance

```
private:
   int ax;
public:
 //write
 constructor to
 initialize ax
 //Write
 destructor
 }
```

```
class B:public A {
   private:
      int bx;
   public:
    //write
    constructor to
    initialize bx
    //Write
    destructor

   }
```

```
class C:public B {
  private:
      int cx;
   public:
     //write
     constructor to
     initialize cx
//Write method to
sum ax, bx and cx
//Write destructor
}
```

```
int main(){
 C c;
//call
//methods of
//class C
return 0
}
```

```cpp
#include <bits/stdc++.h>
using namespace std;

class A {
protected:
    int ax;
public:
    A() {
        ax = 10;
        cout << "A constructor\n";
    }
    ~A() {
        cout << "A destructor\n";
    }
};

class B : public A {
protected:
    int bx;
public:
    B() {
        bx = 20;
        cout << "B constructor\n";
    }
    ~B() {
        cout << "B destructor\n";
    }
};

class C : public B {
private:
    int cx;
public:
    C() {
        cx = 30;
        cout << "C constructor\n";
    }
    void sum() {
        cout << "Sum = " << ax + bx + cx << endl;
    }
    ~C() {
        cout << "C destructor\n";
    }
};

int main() {
    C c;
    c.sum();
```

```
    return 0;
}
```

Multiple inheritance

private:
int ax;
public:
//write
constructor
to initialize
ax
//Write
destructor
}
	class B{
	private:
	int bx;
	public:
	//write
	constructor
	to initialize
	bx
	//Write
	destructor
	}
		class C:public A,
		Public B{
		private:
		int cx;
		public:
		//write
		constructor to
		initialize cx
		//Write method to sum
		ax, bx and cx
		//Write destructor
		}
			int main(){
			C c;
			//call
			//methods of
			//class C
			return 0
			}

```
#include <bits/stdc++.h>
using namespace std;

class A {
protected:
    int ax;
public:
    A() {
        ax = 10;
        cout << "A constructor\n";
    }
    ~A() {
        cout << "A destructor\n";
    }
};

class B {
protected:
    int bx;
public:
    B() {
        bx = 20;
        cout << "B constructor\n";
    }
    ~B() {
        cout << "B destructor\n";
    }
};

class C : public A, public B {
private:
```

```
    int cx;
public:
    C() {
        cx = 30;
        cout << "C constructor\n";
    }
    void sum() {
        cout << "Sum = " << ax + bx + cx << endl;
    }
    ~C() {
        cout << "C destructor\n";
    }
};

int main() {
    C c;
    c.sum();
    return 0;
}
```

Heirarchical inheritance

class A{
private:
int ax;
public:
//write
constructor to
initialize ax
//Write
destructor
}
	class B:public A
	{
	private:
	int bx;
	public:
	//write
	constructor
	to initialize
	bx
	//Write
	destructor
	}
		class C:public A {
		private:
		int cx;
		public:
		//write
		constructor to
		initialize cx
		//Write method to
		sum ax, bx and cx
		//Write destructor
		}
			int main(){
			B b;
			C c;
			//call
			//methods of
			//class B & C
			return 0
			}

```
#include <bits/stdc++.h>
using namespace std;

class A {
protected:
    int ax;
public:
    A() {
        ax = 10;
        cout << "A constructor\n";
```

```
    }
    ~A() {
        cout << "A destructor\n";
    }
};

class B : public A {
private:
    int bx;
public:
    B() {
        bx = 20;
        cout << "B constructor\n";
    }
    ~B() {
        cout << "B destructor\n";
    }
};

class C : public A {
private:
    int cx;
public:
    C() {
        cx = 30;
        cout << "C constructor\n";
    }
    void sum() {
        cout << "Sum = " << ax + cx << endl;
    }
    ~C() {
        cout << "C destructor\n";
    }
};

int main() {
    B b;
    C c;
    c.sum();
    return 0;
}
```

Hybrid (Diamond) inheritance [virtual class]

class A{
private:
int ax;
public:
//write
constructo
r to
initialize
ax
//Write
destructor
}
	class
	B:public A {
	private:
	int bx;
	public:
	//write
	constructo
	r to
	initialize
	bx
	//Write
	destructor
	
	}
		class C:public A {
		private:
		int cx;
		public:
		//write
		constructor
		to
		initialize
		cx
		//Write
		destructor
		
		}
			class D:public B,
			public C {
			private:
			int dx;
			public:
			//write
			constructor to
			initialize dx
			//Write method
			to sum ax, bx
			cx and dx
			//Write
			destructor
			}
				int
				main(){
				D d;
				//call
				//methods
				of
				//class D
				return 0
				}

```cpp
#include <bits/stdc++.h>
using namespace std;

class A {
protected:
    int ax;
public:
    A() {
        ax = 10;
        cout << "A constructor\n";
    }
    ~A() {
        cout << "A destructor\n";
    }
};

class B : virtual public A {
protected:
    int bx;
public:
    B() {
        bx = 20;
        cout << "B constructor\n";
    }
    ~B() {
        cout << "B destructor\n";
    }
};

class C : virtual public A {
protected:
    int cx;
public:
```

```
    C() {
        cx = 30;
        cout << "C constructor\n";
    }
    ~C() {
        cout << "C destructor\n";
    }
};

class D : public B, public C {
private:
    int dx;
public:
    D() {
        dx = 40;
        cout << "D constructor\n";
    }
    void sum() {
        cout << "Sum = " << ax + bx + cx + dx << endl;
    }
    ~D() {
        cout << "D destructor\n";
    }
};

int main() {
    D d;
    d.sum();
    return 0;
}
```

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Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering
24 Series

Course No.: CSE 1203
Course Title: Object Oriented Programmming

Experiment No. : 04
Experiment Title : Polymorphism

Submitted by:	**Submitted to:**
Muhammad Ragib Ishrak Roll: 2403013 Section: A	Dr. Md. Shahid Uz Zaman Professor Dept. of Computer Science & Engineering, RUET

Date of Experiment :
Date of Submission : 26 April 2026

Topic 1[Method/Function Overloading]

Problem Statement: Define a class Test where overload a method Sum() to sum numbers sent from main() function.

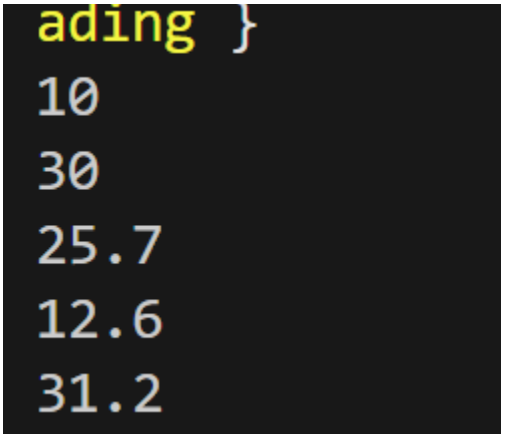
Solution:

```
#include<bits/stdc++.h>
using namespace std;
class Test
{
    public:
    int Sum(int t)
    {
        return t;
    }
    int Sum(int a, int b)
    {
        return a+b;
    }
    double Sum(double a, int b)
    {
        return a+b;
    }
    double Sum(int a, double b)
    {
        return a+b;
    }
    double Sum(double a, double b)
    {
        return a+b;
    }

};
int main()
```

```
{
    Test t;
    cout << t.Sum(10) << endl;
    cout << t.Sum(10,20) << endl;
    cout << t.Sum(5.7,20) << endl;
    cout << t.Sum(10,2.6) << endl;
    cout << t.Sum(10.5,20.7) << endl;
    return 0;
}
```

Output:



Topic 2 [Binary Operator Overloading]: Suppose in a AC circuit, there are 3 impedances z1=3+j4, z2=4-j3 and z3=j6 are connected in parallel. Now find the current in the circuit if input voltage is 100+j50. Implement operator overloading concept for your calculation. Use class Circuit and initialize the impedance values (real & img) by a constructor.

Solution:

```
#include <iostream>
using namespace std;

class Circuit {
private:
```

```
    double real;
    double img;

public:
    Circuit(double r = 0, double i = 0) {
        real = r;
        img = i;
    }
    Circuit operator+(Circuit c) {
        return Circuit(real + c.real, img + c.img);
    }
    Circuit operator/(Circuit c) {
        double denom = c.real * c.real + c.img * c.img;
        double r = (real * c.real + img * c.img) / denom;
        double i = (img * c.real - real * c.img) / denom;
        return Circuit(r, i);
    }
    void display() {
        if (img >= 0)
            cout << real << " + j" << img << endl;
        else
            cout << real << " - j" << -img << endl;
    }
};

int main() {
    Circuit z1(3, 4);
    Circuit z2(4, -3);
    Circuit z3(0, 6);
```

```
    Circuit V(100, 50);
    Circuit one(1, 0);
    Circuit y1 = one / z1;
    Circuit y2 = one / z2;
    Circuit y3 = one / z3;

    Circuit Yeq = y1 + y2 +
y3;

    Circuit Zeq = one / Yeq;

    Circuit I = V / Zeq;

    cout << "Equivalent Impedance Z = ";
    Zeq.display();

    cout << "Circuit Current I = ";
    I.display();

    return 0;
}
```

```
PS D:\Satu\1-2\CSE 1204\week 4> cd "d:\Satu\1
g++ binaryoprtroverloading.cpp -o binaryoprtr
yoprtroverloading }
Equivalent Impedance Z = 2.31193 + j1.70642
Circuit Current I = 38.3333 - j6.66667
```

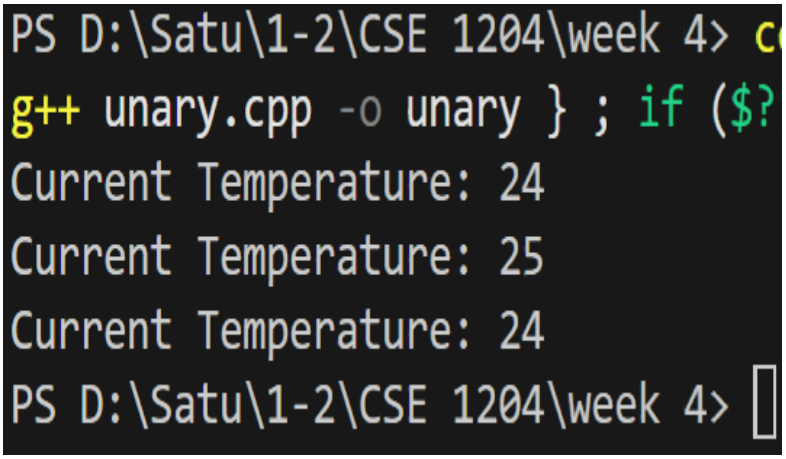
Topic 3 [Unary Operator Overload]

Problem statement: You need to write a program for remote controller of Air-conditioner. You need the control two buttons like temperature up and temperature down using the concept of unary operator overloading. The main() function would be look like:

Solution:

```
#include <iostream>
```

```
using namespace std;
class AirConditioner
{
    int temperature;
public:
    AirConditioner(int t)
    {
        temperature = t;
    }
    void operator++()
    {
        temperature++;
    }
    void operator--()
    {
        temperature--;
    }
    void display()
    {
        cout << "Current Temperature: " << temperature << endl;
    }
};
int main()
{
    AirConditioner ac(24);
    ac.display();
    ++ac;
    ac.display();
    --ac;
    ac.display();
    return 0;
}
```



Topic 4[Method/Function Overriding]

Problem statement: Write a class A with a method Print() and a derived class B with method Print() overloaded. Now observe the output when following statements are written in the main() function-

Solution:

```
#include <iostream>
using namespace std;
class A{
public:
    void Print(){
        cout << "Inside Print() of class A" << endl;
    }
};
class B : public A{
public:
    void Print(){
        cout << "Inside Print() of class B" << endl;
    }
};
int main()
{
    A a;
    a.Print();
    B b;
    b.Print();
    A a2;
    A *p1;
    p1 = &a2;
    p1->Print();
    B b2;
    A *p2;
    p2 = &b2;
    p2->Print();
    return 0;
```

```
PS D:\Satu\1-2\CSE 1204\we
g++ functionoveriding.cpp
ng }
Inside Print() of class A
Inside Print() of class B
Inside Print() of class A
Inside Print() of class A
```

```
}
```

Topic 5 [Friend Function]

Problem Statement: Using the following class, write three friend functions

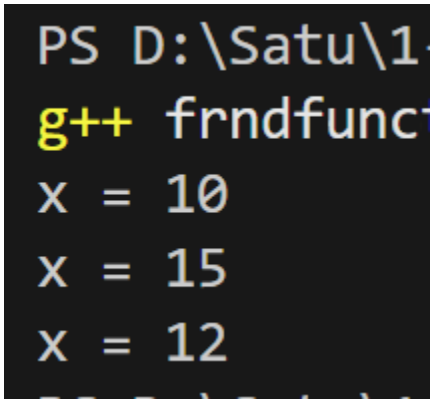
i) Add(): Assign value to the data member x

ii) IncX() : Increase the value of x by m

iii) DecX() : Decrease the value of x by n

Solution:

```
#include <iostream>
using namespace std;
class Sample
{
    int x;
public:
    Sample()
    {
        x = 0;
    }
    friend void Add(Sample &s, int value);
    friend void IncX(Sample &s, int m);
    friend void DecX(Sample &s, int n);
    void display()
    {
        cout << "Value of x: " << x << endl;
    }
};
void Add(Sample &s, int value)
{
    s.x = value;
```



```
}

void IncX(Sample &s, int m)
{
    s.x = s.x + m;
}
void DecX(Sample &s, int n)
{
    s.x = s.x - n;
}
int main()
{
    Sample obj;
    Add(obj, 10);
    obj.display();
    IncX(obj, 5);
    obj.display();
    DecX(obj, 3);
    obj.display();
    return 0;
}
```

Topic 6[Pure Virtual Function]

Problem statement: Modify the class defined in Topic 3 executes the following

```
#include <iostream>
using namespace std;

class A{
public:
```

```
    virtual void Print() = 0;
};

class B : public A{
public:
    void Print(){
        cout << "Inside Print() of class B" << endl;
    }
};
```

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24 Series

Course No.: CSE 1203
Course Title: Object Oriented Programmming

Experiment No. : 05
Experiment Title : Advanced Topic

Submitted by:	**Submitted to:**
Muhammad Ragib Ishrak Roll: 2403013 Section: A	Shyla Afroge Associate Professor Dept. of Computer Science & Engineering, RUET

Date of Experiment :
Date of Submission : 26 April 2026

Module 5 [Advanced Topics]:

Topic 1[Exception Handling]

Problem Statement: In the following input `i` is the index of array `ax[].` The program prints `ax[i].` Then write catch block if i is out of range of `ax[].` Write three `catch` blocks to fullfill the purpose

i) a `catch` block receives the value of `i`
ii) a `catch` block receives string "`Out of Range Error`"
iii) a default `catch()` if above two `catch` block doesn't match

```
#include <iostream>
using namespace std;

int main()
{
  int i;
  int ax[5]={10,20,60,40,30};
  cout<<"enter index:";
  cin>>i;
  cout<<"ax["<<i<<"]="<<ax[i]<<endl;
}
```

Code:

```
#include <iostream>
using namespace std;

int main()
{
  int i;
  int ax[5]={10,20,60,40,30};
  cout<<"Enter index(int):";
  cin>>i;
  try
  {
    if(i>4 || i<0)
    {
        throw(i);
    }
    if(cin.fail())
    {
        throw("notinteger");
    }
    cout<<"ax["<<i<<"]="<<ax[i]<<endl;
  }
  catch(int i)
{
    cout << "Out of Range" << endl;
}
  catch(const char *s)
```

```
  {
    cout << s << endl;
  }
  catch(...)
  {
    cout << "Exception is exception" << endl;
  }
  return 0;
}
```

Topic 2 [**class Template**]:

Problem Statement: In the following class data members `x` and `y` are integers and the method `Sum()` adds `x` and `y`. However, we need to perform the sum of `int+int, int+double, douuble+int` and `double+double` . To achieve it, change the definition of `x,y,setData()` and `Sum()` accordingly.

```
class A{
 private:
    int x;
    int y;
  public:
    void setData(int x,int y){
     this->x=x;
     this->y=y;
    }
    int Sum(){
      int s;
      s=x+y;
      return s;
    }
};
int main(){
 //write required statements to call SetData() & Sum()
}
```

Code:

```
#include <iostream>
using namespace std;
template <class T1, class T2>
class A
{
```

```
    T1 x;
    T2 y;
public:
    void setData(T1 x, T2 y) {
       this-> x=x;
        this ->y =y;
    }
    auto Sum()
    {
        return x + y;
    }
};
int main()
{

    A<int, int> obj1;
    obj1.setData(10, 20);
    cout << "Sum(int, int): " << obj1.Sum() << endl;
    A<int, double> obj2;
    obj2.setData(10, 5.5);
    cout << "Sum(int, double): " << obj2.Sum() << endl;
    A<double, int> obj3;
    obj3.setData(3.5, 7);
    cout << "Sum(double, int): " << obj3.Sum() << endl;
    A<double, double> obj4;
    obj4.setData(2.5, 3.5);
    cout << "Sum(double, double): " << obj4.Sum() << endl;
    return 0;
}
```

Topic 3 [STL:Array class]

Problem statement: Declare a STL array object ax with 6 elements and do the following:

i) Assign 10,60,30,70,20 to ax using single statement
ii) Print third element of ax using at() function
iii) Print first element of ax using front() function
iv) Print last element of ax using back() function
v) Fill the elements of ax using fill() function
vi) Test whether ax is empty or not using empty() function
vii) Print size of ax
viii) Print maximum size of ax using max_size() function
ix) Print address of first element of ax using begin() function
x) Print address of last element of ax using end() function

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;
int main(){
  array<int,6>ax;
  //write statements
}
```

Code:

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;

int main() {
    array<int, 6> ax;

    // i) assign using single statement
    ax = {10, 60, 30, 70, 20, 90};

    // ii) third element
    cout << "Third element: " << ax.at(2) << endl;

    // iii) first element
    cout << "First element: " << ax.front() << endl;

    // iv) last element
    cout << "Last element: " << ax.back() << endl;

    // v) fill
    ax.fill(5);
    cout << "After fill(): ";
    for (int x : ax) cout << x << " ";
    cout << endl;
```

```
    // vi) empty
    cout << "Is empty: " << (ax.empty() ? "Yes" : "No") << endl;

    // vii) size
    cout << "Size: " << ax.size() << endl;

    // viii) max_size
    cout << "Max size: " << ax.max_size() << endl;

    // ix) address of first element
    cout << "Address of first element: " << &(*ax.begin()) << endl;

    // x) address of last element
    cout << "Address of last element: " << &(*(ax.end() - 1)) << endl;

    return 0;
}
```

```
PS D:\Satu\1-2\CSE 1204\week 5> cd "d:\s
g++ stlarray.cpp -o stlarray } ; if ($?)
Third element: 30
First element: 10
Last element: 90
After fill(): 5 5 5 5 5 5
Is empty: No
Size: 6
Max size: 6
Address of first element: 0x61fee4
Address of last element: 0x61fef8
```

Topic 4[STL: pair class]

Problem statement: Define a pair class object px with int and string elements. Write statements to do the following

i) Assign 10 to int and “Rajshahi” to px using make_pair() function
ii) Print int data member by first
iii) Print string data member by second
iv) Modify first data member to 20 using get<>() function
v) Declare another pair bx and assign values to bx and swap it with ax

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;
int main(){
  pair<int,string>px;
  //write statements
}
```

Code:

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;
```

```cpp
int main() {
    pair<int, string> px;

    // i)
    px = make_pair(10, "Rajshahi");

    // ii)
    cout << "First: " << px.first << endl;

    // iii)
    cout << "Second: " << px.second << endl;

    // iv)
    get<0>(px) = 20;
    cout << "Modified First: " << px.first << endl;

    // v)
    pair<int, string> bx;
    bx = make_pair(50, "Dhaka");

    px.swap(bx);

    cout << "After swap, px = (" << px.first << ", " << px.second << ")" << endl;
    cout << "After swap, bx = (" << bx.first << ", " << bx.second << ")" << endl;

    return 0;
}
```

```
PS D:\Satu\1-2\CSE 1204\week 5>
g++ stlpair.cpp -o stlpair } ;
First: 10
Second: Rajshahi
Modified First: 20
After swap, px = (50, Dhaka)
After swap, bx = (20, Rajshahi)
```

Topic 5 [STL: tuple class]

iii) Print string data member by get() function
iv) Print double data member by get() function
v) Modify third data member to 3.7 using get<>() function
vi) Declare another tuple bx and assign values to bx and swap it with ax

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;
int main(){
  tuple<int,string,double>tx;
  //write statements
}
```

Code:

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;

int main() {
    tuple<int, string, double> tx;

    // i)
    tx = make_tuple(100, "Kamal", 3.5);

    // ii)
    cout << "Int: " << get<0>(tx) << endl;

    // iii)
    cout << "String: " << get<1>(tx) << endl;

    // iv)
    cout << "Double: " << get<2>(tx) << endl;

    // v)
    get<2>(tx) = 3.7;
    cout << "Modified Double: " << get<2>(tx) << endl;

    // vi)
    tuple<int, string, double> bx;
    bx = make_tuple(200, "Rahim", 4.2);

    tx.swap(bx);

    cout << "After swap, tx = ("
         << get<0>(tx) << ", "
         << get<1>(tx) << ", "
         << get<2>(tx) << ")" << endl;

    cout << "After swap, bx = ("
```

```
        << get<0>(bx) << ", "
        << get<1>(bx) << ", "
        << get<2>(bx) << ")" << endl;

    return 0;
}
```

```
PS D:\Satu\1-2\CSE 1204\week 5> 
g++ stlpair.cpp -o stlpair } ; i
First: 10
Second: Rajshahi
Modified First: 20
After swap, px = (50, Dhaka)
After swap, bx = (20, Rajshahi)
```

Topic 6 [STL: stack class]

Problem statement: Write a program to create and manipulate Stack using `stack` class and Perform the following operations using the specified method.

i) use `push()` method to push data
ii) use `pop()` method to pop data
iii) use `top()` method to display top element
iv) use `empty()` method to check whether stack is empty or not

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;
int main(){
  stack<int>st;
  //write statements
}
```

Code:

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;

int main() {
    stack<int> st;

    // push()
    st.push(10);
    st.push(20);
    st.push(30);

    // top()
    cout << "Top element: " << st.top() << endl;

    // pop()
    st.pop();
    cout << "After pop, top element: " << st.top() << endl;

    // empty()
    if (st.empty())
        cout << "Stack is empty" << endl;
    else
        cout << "Stack is not empty" << endl;

    return 0;
}
```

```
PS D:\Satu\1-2\CSE 1204\wee
g++ stackclass.cpp -o stack
Top element: 30
After pop, top element: 20
Stack is not empty
```

Topic 7 [STL: queue class]

Problem statement: Write a program to create and manipulate Queue using queue

class. Perform the following operations using the specified method.

i) use `push()` method to push data
ii) use `pop()` method to pop data
iii) use `front()` method to display front element
iv) use `back()` method to display rear element
v) use `empty()` method to check whether queue is empty or not

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;
int main(){
  queue<int>qu;
```

Code:

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;

int main() {
    queue<int> qu;

    // push()
    qu.push(10);
    qu.push(20);
    qu.push(30);

    // front()
    cout << "Front element: " << qu.front() << endl;

    // back()
    cout << "Rear element: " << qu.back() << endl;

    // pop()
    qu.pop();
    cout << "After pop, front element: " << qu.front() << endl;

    // empty()
    if (qu.empty())
        cout << "Queue is empty" << endl;
    else
        cout << "Queue is not empty" << endl;

    return 0;
}
```

```
  //write statements
}
```

Topic 8 [STL: list class]

Problem statement: Write a program to create and manipulate linked list using `list` class and its following methods to

i) insert 8 integers using **`push_back()`** method
ii) insert two elements using **`push_front()`** method
iii) display all the elements of the list in forward direction with user-defined Display() method using **`begin()`** and **`end()`** methods and iterator
iv) display all the elements of the list in reverse direction with a user-defined DisplayRev() method using **`rbegin()`** and rend() method and iterator
v) display front element using **`front()`** method
vi) display back element using **`back()`** method
vii) delete front element using **`pop_back()`** method

viii) delete front element using **pop_front()** method
ix) search an element **x** using **find()** method
x) insert a new element **x** before an existing element **y** using **insert()** method
xi) insert a new element **x** after an existing element **y** using **insert()** method
xii) count a particular element **x**
xiii) count a elements with condition using predicate function
xiv) delete a particular element **x** with **erase()** method
xv) delete first 4 elements with **erase()** method
xvi) delete a particular element **x** with **remove()** method
xvii) delete a elements with condition using **remove_if()** method using predicate function
xviii) assign elements from another list using **assign()** method
xix) assign elements from an array using **assign()** method
xx) sort the list using **sort()** method
xxi) Delete consecutive similar elements using **unique()** method

Code:

```
#include <iostream>
#include <bits/stdc++.h>
#include <iterator>
#include <algorithm>
using namespace std;

int main(){
  list<int>li;
  //write statements
}
```

```
#include <bits/stdc++.h>
#include <iterator>
#include <algorithm>
using namespace std;

bool isEven(int x) {
    return x % 2 == 0;
}

void Display(list<int> li) {
    list<int>::iterator it;
    for (it = li.begin(); it != li.end(); it++) {
        cout << *it << " ";
    }
    cout << endl;
}

void DisplayRev(list<int> li) {
    list<int>::reverse_iterator it;
```

```
    for (it = li.rbegin(); it != li.rend(); it++) {
        cout << *it << " ";
    }
    cout << endl;
}

int main() {
    list<int> li;

    // i) insert 8 integers using push_back()
    li.push_back(10);
    li.push_back(20);
    li.push_back(30);
    li.push_back(40);
    li.push_back(50);
    li.push_back(60);
    li.push_back(70);
    li.push_back(80);

    // ii) insert two using push_front()
    li.push_front(5);
    li.push_front(1);

    cout << "Forward: ";
    Display(li);

    // iv) reverse display
    cout << "Reverse: ";
    DisplayRev(li);

    // v) front()
    cout << "Front element: " << li.front() << endl;

    // vi) back()
    cout << "Back element: " << li.back() << endl;

    // vii) delete back using pop_back()
    li.pop_back();
    cout << "After pop_back: ";
    Display(li);

    // viii) delete front using pop_front()
    li.pop_front();
    cout << "After pop_front: ";
    Display(li);

    // ix) search an element x using find()
    int x = 30;
    auto it = find(li.begin(), li.end(), x);
```

```
    if (it != li.end())
        cout << x << " found" << endl;
    else
        cout << x << " not found" << endl;

    // x) insert a new element x before existing y
    int newVal1 = 25, y1 = 30;
    auto it1 = find(li.begin(), li.end(), y1);
    if (it1 != li.end()) {
        li.insert(it1, newVal1);
    }
    cout << "After insert before 30: ";
    Display(li);

    // xi) insert a new element x after existing y
    int newVal2 = 35, y2 = 30;
    auto it2 = find(li.begin(), li.end(), y2);
    if (it2 != li.end()) {
        ++it2;
        li.insert(it2, newVal2);
    }
    cout << "After insert after 30: ";
    Display(li);

    // xii) count a particular element x
    cout << "Count of 30: " << count(li.begin(), li.end(), 30) << endl;

    // xiii) count elements with condition
    cout << "Count of even elements: " << count_if(li.begin(), li.end(), isEven) <<
endl;

    // xiv) delete a particular element x with erase()
    auto it3 = find(li.begin(), li.end(), 25);
    if (it3 != li.end()) {
        li.erase(it3);
    }
    cout << "After erase(25): ";
    Display(li);

    // xv) delete first 4 elements with erase()
    auto start = li.begin();
    auto finish = li.begin();
    advance(finish, 4);
    li.erase(start, finish);
    cout << "After deleting first 4 elements: ";
    Display(li);

    // xvi) delete a particular element x with remove()
    li.remove(60);
```

```
    cout << "After remove(60): ";
    Display(li);

    // xvii) delete elements with condition using remove_if()
    li.remove_if(isEven);
    cout << "After remove_if(isEven): ";
    Display(li);

    // xviii) assign elements from another list
    list<int> li2;
    li2.push_back(100);
    li2.push_back(200);
    li2.push_back(300);

    li.assign(li2.begin(), li2.end());
    cout << "After assign from another list: ";
    Display(li);

    // xix) assign elements from an array
    int arr[] = {7, 8, 9, 10};
    li.assign(arr, arr + 4);
    cout << "After assign from array: ";
    Display(li);

    // xx) sort the list
    li.push_back(2);
    li.push_back(15);
    li.sort();
    cout << "After sort: ";
    Display(li);

    // xxi) delete consecutive similar elements using unique()
    li.push_back(15);
    li.push_back(15);
    li.push_back(20);
    li.push_back(20);
    li.sort();
    cout << "Before unique: ";
    Display(li);

    li.unique();
    cout << "After unique: ";
    Display(li);

    return 0;
}
```

```
After erase(25): 5 10 20 30 35 40 50 60 70
After deleting first 4 elements: 35 40 50 60 70
After remove(60): 35 40 50 70
After remove_if(isEven): 35
After assign from another list: 100 200 300
After assign from array: 7 8 9 10
After sort: 2 7 8 9 10 15
Before unique: 2 7 8 9 10 15 15 15 20 20
After unique: 2 7 8 9 10 15 20 _
```

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Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering
24 Series

Course No.: CSE 1203
Course Title: Object Oriented Programmming

Experiment No. : 06
Experiment Title : Data Structure with STL

Submitted by:	**Submitted to:**
Muhammad Ragib Ishrak Roll: 2403013 Section: A	Shyla Afroge Associate Professor Dept. of Computer Science & Engineering, RUET

Date of Experiment :
Date of Submission : 26 April 2026

Module 06 [Data Structure with STL]
(for Week 7)

Topic 1 [STL: stack class]

Problem statement: Write a menu program to create and manipulate Stack using `stack` class and perform the following operations using the specified method.

i) use `push()` method to push data
ii) use `pop()` method to pop data
iii) use `top()` method to display top element
iv) use `empty()` method to check whether stack is empty or not

```
** Stack **
1. Push
2. Pop
3. Display()
4. Exit
   Enter your option:
```

Code:

```
#include<bits/stdc++.h>
using namespace std;

int main(){
    stack<int> st;
    int opt, val;

    while(true){
        cout<<"\n1.Push\n2.Pop\n3.Top\n4.Check Empty\n5.Exit\n";
        cin>>opt;

        switch(opt){
            case 1:
                cout<<"Enter value: ";
                cin>>val;
                st.push(val);
                break;

            case 2:
                if(!st.empty()){
                    cout<<"Popped: "<<st.top()<<endl;
                    st.pop();
                } else cout<<"Stack Empty\n";
                break;

            case 3:
                if(!st.empty())
                    cout<<"Top: "<<st.top()<<endl;
```

```
                else cout<<"Stack Empty\n";
                break;

            case 4:
                cout<<(st.empty()?"Empty":"Not Empty")<<endl;
                break;

            case 5:
                return 0;
        }
    }
}
```

Topic 2 [STL: queue class]

Problem statement: Write a menu program to create and manipulate Queue using queue class. Perform the following operations using the specified method.

i) use `push()` method to enqueue data
ii) use `pop()` method to dequeue data
iii) use `front()` method to display front element
iv) use `back()` method to display rear element
v) use `empty()` method to check whether queue is empty or not

```
** Queue **
1. Enqueue
2. Dequeue
3. Display Front
4. Display Rear
5. Exit
   Enter your option:
```

Code:

```
#include<bits/stdc++.h>
using namespace std;

int main(){
    queue<int> q;
    int opt, val;

    while(true){
        cout<<"\n1.Enqueue\n2.Dequeue\n3.Front\n4.Rear\n5.Exit\n";
        cin>>opt;

        switch(opt){
            case 1:
                cin>>val;
                q.push(val);
                break;
```

```
            case 2:
                if(!q.empty()){
                    cout<<"Removed: "<<q.front()<<endl;
                    q.pop();
                } else cout<<"Queue Empty\n";
                break;

            case 3:
                if(!q.empty())
                    cout<<"Front: "<<q.front()<<endl;
                break;

            case 4:
                if(!q.empty())
                    cout<<"Rear: "<<q.back()<<endl;
                break;

            case 5:
                return 0;
        }
    }
}
```

Topic 3 [STL: vector class]

Problem statement: Write a menu program to create and manipulate linked list using `vector` class and use the following methods

i) **`push_back()`**
ii) **`push_front()`**
iii) **`begin()`**
iv) **`end()`**

v) **pop_back()**
vi) **pop_front()**
vii) **insert()**
viii) **erase()**

```
** Linked List **
1. Insert
2. Delete
3. Update
4. Search
5. Exit
   Enter your option:
```

Code:

```
#include<bits/stdc++.h>
using namespace std;

int main(){
    vector<int> v;
    int opt, val, pos;

    while(true){
        cout<<"\n1.Insert\n2.Delete\n3.Update\n4.Search\n5.Exit\n";
        cin>>opt;

        switch(opt){
            case 1:
                cin>>val;
                v.push_back(val);
                break;

            case 2:
                cin>>pos;
                if(pos>=0 && pos<v.size())
                    v.erase(v.begin()+pos);
                break;

            case 3:
                cin>>pos>>val;
                if(pos>=0 && pos<v.size())
                    v[pos]=val;
                break;

            case 4:
                cin>>val;
                for(int i=0;i<v.size();i++)
                    if(v[i]==val)
                        cout<<"Found at "<<i<<endl;
                break;
```

```
            case 5:
                return 0;
        }
    }
}
```

Topic 3.1 [STL: vector class]: Add more feature to the menu of Topic 3.

Module 07 [UML+Java Basics]
(for Week 8: 19-23/7/2025)

Topic 1 [UML]

Problem statement: Draw the UML diagram for the following class

```
class Test{
  private: int x;
  protected: int y;
  public: int z;
  public:
    void SetData(int a,int b,int c){
      x=a;y=b;z=c;
  }
  float getAverage(){
   Return (x+y+z)/3.0;
 }
```

Code:

| Test |

| - x : int |

| # y : int |

| + z : int |

| + SetData(a:int, |

| b:int, |

| c:int) |

| + getAverage():float |

Topic 2 [UML: Aggregation]

Problem statement: Create two classes Doctor and Patient to store the information of doctors and patients respectively. Now write codes to create association between these two classes by aggregation.

[note: if Patient class is destroyed then Doctor class is still exist]

class Doctor{
//write your code
}
	class Patient{
	//write your code
	}
		int main(){
		//write your code
		}

Code:

```
#include<bits/stdc++.h>
using namespace std;
class Student
{
    public:
    str#include<bits/stdc++.h>
using namespace std;

class Patient{
public:
    string name;
    Patient(string n){ name = n; }
};

class Doctor{
public:
    Patient* p; // aggregation (reference)

    Doctor(Patient* p){
        this->p = p;
    }

    void show(){
        cout<<"Patient: "<<p->name<<endl;
    }
};

int main(){
    Patient p("Ragib");
    Doctor d(&p);

    d.show();
}ing name;
    Student(string n)
    {
        name = n;
    }
```

```
};
```

Topic 3 [UML: Composition]

Problem statement: Create two classes Doctor and Patient to store the information of doctors and patients respectively. Now write codes to create association between these two classes by composition.

[note: if Doctor class is destroyed then Patient class is also destroyed]

class Doctor{
//write your code
}
	class Patient{
	//write your code
	}
		int main(){
		//write your code
		}

Code:

```
#include<bits/stdc++.h>
using namespace std;

class Patient{
public:
    string name;
    Patient(string n){ name=n; }
};

class Doctor{
public:
    Patient p; // composition

    Doctor(string name): p(name){}

    void show(){
        cout<<"Patient: "<<p.name<<endl;
    }
};

int main(){
    Doctor d("Ragib");
    d.show();
}
```

Topic 4: [**java Programming Basics**]

Problem Statement: Write java programs to do the following

i) Display your name and address in two lines

ii) Take two integers as inputs and prints the bigger one

iii) Input a series of 10 numbers into an array ax[] and finds the largest, smallest and average of these numbers. Also write code for searching and sorting.

Code(i):

```
class A{
```

```
    public static void main(String[] args){
        System.out.println("Ragib Ishrak");
        System.out.println("Bangladesh");
    }
}
```

```
PS D:\Satu\1-2\CSE
" ; if ($?) { javac
Ragib Ishrak
Bangladesh
```

Code(ii):

```
import java.util.*;
class A{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int a=sc.nextInt(), b=sc.nextInt();
        System.out.println(Math.max(a,b));
    }
}
```

```
1 error
PS D:\Satu\1-2\CSE 120
($?) { javac A.java }
17 45
45
```

Code(iii):

```
import java.util.*;
class A{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int a=sc.nextInt(), b=sc.nextInt();
        System.out.println(Math.max(a,b));
    }
}
```

```
PS D:\Satu\1-2\CSE 1204\labmodule6> cd "d:\Satu\1-2\CSE 1204\labmodule6\" ; if
($?) { javac main.java } ; if ($?) { java main }
Error: Could not find or load main class main
Caused by: java.lang.ClassNotFoundException: main
PS D:\Satu\1-2\CSE 1204\labmodule6>
```

Topic 5 [java Programming private method]

Problem Statement: Verify that private methods can be used only inside the class but it cannot be used from outside the class.

Code:

```
class A{
    private void show(){
        System.out.println("Private method");
    }

    void call(){
        show(); // allowed
    }

    public static void main(String[] args){
        A obj=new A();
        obj.call();
        // obj.show(); ❌ not allowed
    }
}
```

Topic 6 [**java Programming static method/member**]
Problem Statement: Verify that static methods can access static method/member but cannot access to non-static method/member.

Code:

```
class A{
    static int x=10;
    int y=20;

    static void show(){
        System.out.println(x); // allowed
        // System.out.println(y); ❌ not allowed
    }

    public static void main(String[] args){
        show();
    }
}
```

Topic 7 [**java Programming**]
Problem Statement: Write a Java Program to initialize radius of a circle using constructor/Setter and calculate area of the circle.

Code:

```
class Circle{
    double r;

    Circle(double r){
        this.r=r;
    }

    double area(){
        return Math.PI*r*r;
    }

    public static void main(String[] args){
        Circle c=new Circle(5);
        System.out.println("Area="+c.area());
    }
}
```

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Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering
24 Series

Course No.: CSE 1203
Course Title: Object Oriented Programmming

Experiment No. : 07
Experiment Title : Java Basics

Submitted by:	**Submitted to:**
Muhammad Ragib Ishrak Roll: 2403013 Section: A	Shyla Afroge Associate Professor Dept. of Computer Science & Engineering, RUET

Date of Experiment :
Date of Submission : 26 April 2026

Module 07 [Java Basics]

Topic 1 [**java Programming private method**]

Problem Statement: Verify that private methods can be used only inside the class but it cannot be used from outside the class.

Code:

```
class A
{
    private void show()
    {
        System.out.println("Private Method");
    }
    public void display()
    {
        System.out.println("Withen class");
        show();
    }

}
public class Topic1
{
    public static void main(String[] args)
    {
        A a = new A();
        a.display();
        //a.show();
    }
}
```

Topic 2 [**java Programming static method/member**]

Problem Statement: Verify that static methods can access static method/member but cannot access to non-static method/member.

Code:

```
class A
{
    public static int x = 10;
    int y = 20;
    void display()
```

```
    {
        System.out.println("Non static: "+ y);
    }

    public static void show()
    {
        System.out.println("Static method: "+x);
    }
}
public class Topic2
{
    public static void main(String[] args)
    {
        A a = new A();
        a.display();
        a.show();
    }

}
```

Topic 3 [**java Programming: Anonymous object**]

Problem Statement: Write a Java program that creates a Point object using anonymous object using the following getPoint() method.

```
Point pt=getPoint();
```

Code:

```
 class Point {
    int x;
    int y;

    Point(int x, int y) {
        this.x = x;
        this.y = y;
    }

    void showPoint() {
        System.out.println("X = " + x + ", Y = " + y);
    }
}

public class Mainpoint {

    static Point getPoint() {
        return new Point(5, 10);
    }

    public static void main(String[] args) {
```

```
        getPoint().showPoint();
    }
}
```

Topic 4 [**java Programming: ArrayList<> object**]

Problem Statement: Write a Java program performs the following operations using ArrayList Object

i) Declare a ArrayList object ax to store flowers
ii) Add three flowers: Rose, Lilie, Carnation, Daisie, and Tulip
iii) Insert a new flower Peonie in the second position of ax
iv) Delete the flower Lilie
v) Find any flower in ax
vi) Sort the list
vii) Delete the list

Code:

```
import java.util.*;
public class Topic2
{
    public static void main(String[] args)
    {
    ArrayList<String> ax = new ArrayList<>();
    ax.add("Rose");
    ax.add("Lilie");
    ax.add("Carnation");
    ax.add("Daisie");
    ax.add("Tulip");

    System.out.println("Initial Array: "+ax);

    ax.add(1,"Peonie");
    System.out.println("After Insertion: "+ ax);

    ax.remove("Lilie");
    System.out.println("After Deletion: "+ ax);

    String search = "Tulip";
    if(ax.contains(search))
    {
        System.out.println(search+"found in the list");
    }
    else
    {
        System.out.println(search + "not found");
    }
```

```
    Collections.sort(ax);
    System.out.println("Sorted list"+ ax);

    ax.clear();
    System.out.println("After Clearing: "+ ax);
    }

}
```

Topic 5 [inheritance: use of super keyword]

Problem Statement: For the following Java program write

i) to display x of class A in class B

ii) statement to call getX() of class A in class B

iii) to call parameterized constructor of in class B

`class A{`
`  int x;`
`  public A(){`
`  x=0;`
`  }`
`  public A(int x){`
`  this.x=x;`
`  }`	

`  public int getX(){`
`  return(x+10);`
`  }`
 	
`class B extends A{`
`  int x=20;`
`  public int getX(){`
`  return(x+10);`
`  }`
`}`
`public class First{`
`  public static void main(String[] args) {`
 	
`  }`
`}`	

Code:

```
class A
{
    int x;
    public A()
    {
        x=0;
    }
    public A(int x)
```

```
    {
    this.x=x;
    }
    public int getX()
    {
    return(x+10);
    }
}
class B extends A
{
    int x=20;
    public int getX()
    {
    return(x+10);
    }
    public int getXa()
    {
        return super.x;

    }
    public void show()
    {
        System.out.println(super.getX());
    }
    B(int x)
    {
        super(x);
        System.out.println("Super Parameterized Constructor");

    }
}
public class First
{
public static void main(String[] args)
{

    B b = new B(10);
    System.out.println(b.getXa());
    System.out.println(b.getX());
    b.show();
}
}
```

Topic 6 [interface in Java]

Problem Statement: Write three interfaces in Java namely AI,BI and CI with the following activities:

i) Declare method PrintA() in AI
ii) Declare method PrintB() in BI
iii) Declare method PrintC() in CI
iv) write body of PrintA() in class A
v) write body of PrintB() in class B which is inherited from class A
vi) write body of PrintC() in class C which is inherited from class B
vii) Create object of class C in main() of First class and invoke PrintA(), PrintB() and PrintC()

Code:

```
interface Al
{
     void printA();
}
interface Bl
{
    void printB();


}
interface Cl
{
    void printC();


}
class A implements Al
{
    @Override
    public void printA()
    {
        System.out.println("Print A");
    }

}
class B extends A implements Bl
{
    public void printB()
    {
        System.out.println("Print B");
    }

}
class C extends B implements Cl
{
    public void printC()
    {
        System.out.println("Print C");
```

```
    }
}
public class interfacejava
{
    public static void main(String[] args)
    {
    C obj = new C();
    obj.printA();
    obj.printB();
    obj.printC();
    }

}
```

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24 Series

Course No.: CSE 1203
Course Title: Object Oriented Programmming

Experiment No. : 08
Experiment Title : Advance Topics: Java Programming

Submitted by:	**Submitted to:**
Muhammad Ragib Ishrak Roll: 2403013 Section: A	Shyla Afroge Associate Professor Dept. of Computer Science & Engineering, RUET

Date of Experiment :
Date of Submission : 26 April 2026

Module 08 [Advanced Topics: Java Programming]:
(for Week 11:28-1/3/2026)

Topic 01 [**java Programming: Anonymous object**]

Problem Statement: Write a Java program that creates a Point object using anonymous object using the following getPoint() method.

```
Point pt=getPoint();
```

Code:

```
class Point
{
    int x;
    int y;
    Point(int x, int y)
    {
        this.x=x;
        this.y=y;
    }
    public static Point getPoint(int x, int y)
    {

        return new Point(x, y);

    }
    public void display()
    {
        System.out.println("Point("+x+","+y+")");
    }
}
public class Topic01 {
    public static void main(String[] args) {
        Point pt = Point.getPoint(5,10);
        pt.display();

    }

}
```

Topic 02 [**java Programming: ArrayList<> object**]

Problem Statement: Write a Java program performs the following operations using ArrayList Object

i) Declare a ArrayList object ax to store flowers

ii) Add three flowers: Rose, Lilie, Carnation, Daisie, and Tulip

iii) Insert a new flower Peonie in the second position of ax
iv) Delete the flower Lilie
v) Find any flower in ax
vi) Sort the list
vii) Delete the list

Code:

```
import java.util.*;
public class Topic2
{
    public static void main(String[] args)
    {
    ArrayList<String> ax = new ArrayList<>();
    ax.add("Rose");
    ax.add("Lilie");
    ax.add("Carnation");
    ax.add("Daisie");
    ax.add("Tulip");

    System.out.println("Initial Array: "+ax);

    ax.add(1,"Peonie");
    System.out.println("After Insertion: "+ ax);

    ax.remove("Lilie");
    System.out.println("After Deletion: "+ ax);

    String search = "Tulip";
    if(ax.contains(search))
    {
        System.out.println(search+"found in the list");
    }
    else
    {
        System.out.println(search + "not found");
    }

    Collections.sort(ax);
    System.out.println("Sorted list"+ ax);

    ax.clear();
    System.out.println("After Clearing: "+ ax);
    }

}
```

Topic 03 [inheritance: use of super keyword]

Problem Statement: For the following Java program write

i) to display x of class A in class B

ii) statement to call getX() of class A in class B

iii) to call parameterized constructor of in class B

class A{
  int x;
  public A(){
  x=0;
  }
  public A(int x){
  this.x=x;
  }
 	
  public int getX(){
  return(x+10);
  }
 	
class B extends A{
  int x=20;
  public int getX(){
  return(x+10);
  }
}
public class First{
  public static void main(String[] args) {	

} }	

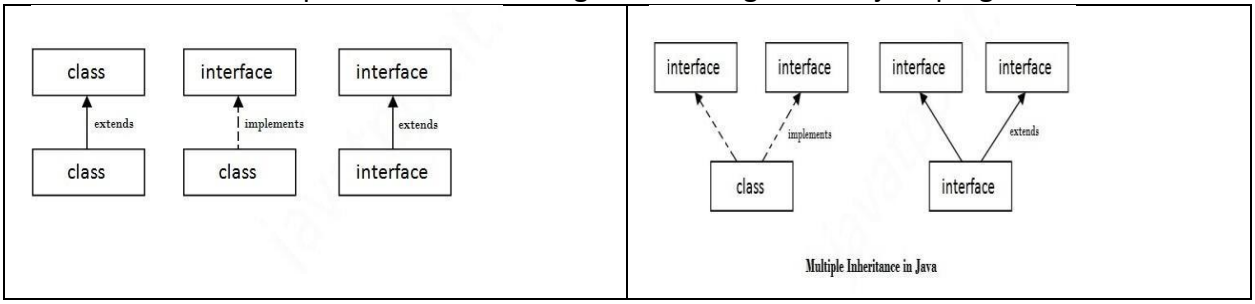
Code:

```
class A
{
    int x;
    public A()
    {
        x=0;
    }
    public A(int x)
    {
    this.x=x;
    }
    public int getX()
    {
    return(x+10);
    }
```

```
}
class B extends A
{
    int x=20;
    public int getX()
    {
    return(x+10);
    }
    public int getXa()
    {
        return super.x;

    }
    public void show()
    {
        System.out.println(super.getX());
    }
    B(int x)
    {
        super(x);
        System.out.println("Super Parameterized Constructor");

    }
}
public class First
{
public static void main(String[] args)
{

    B b = new B(10);
    System.out.println(b.getXa());
    System.out.println(b.getX());
    b.show();
}
}
```

Topic 04 [**inheritance**]

Problem Statement: Implement the following relation diagram into java program



Code:

```
class A
{
    A()
    {
        System.out.println("A created");
    }


}
class B extends A
{
    //super(A);

}
public class Topic41 {
    public static void main(String[] args)
    {
        A b = new B();

    }

}

interface A
{
    void show();


}
class B implements A
{
    @Override
    public void show()
    {
        System.out.println("Implemnetd from A");
    }

}

public class Topic42 {
    public static void main(String[] args) {
        B a = new B();
        a.show();
    }

}

interface a
{
```

```
    void show();
}
interface b extends a {
    @Override
    default void show()
    {
        System.out.println("extened from a");


    }
}
public class Topic43 {
    public static void main(String[] args) {
        b obj = new b() {};
        obj.show();
    }

}

interface A{
    void show();
}
interface B{
    void display();
}
class C implements A,B
{
    @Override
    public void show()
    {
        System.out.println("From a");

    }
    @Override
    public void display()
    {
        System.out.println("From b");
    }
}
public class topic44 {
    public static void main(String[] args)
    {
        C c = new C();
        c.show();
        c.display();

    }

}
```

```java
interface A
{
    void showA();
}
interface B
{
    void showB();
}
interface C extends A,B
{
    @Override
    default void showA()
    {
        System.out.println("From A");


    }
    @Override
    default void showB()
    {
        System.out.println("From B");
    }
}
public class topic45 {
    public static void main(String[] args)
    {
        C c = new C(){};
        c.showA();
        c.showB();
    }

}
```

Topic 05 [**Singleton class**]

Problem Statement: Write a Java program to create and operate a singleton class.

```java
class Singleton
{
    public static int count=0;
    private static Singleton[] instance= new Singleton[2];
    private Singleton()
    {
        count++;

    }
```

```
    public static Singleton getInstance()
    {
       if(count < 2)
       {
        instance[count] = new Singleton();
       }
        return instance[count];
    }
    public void show()
    {
        System.out.println("Singleton instance");
    }
}
public class Topic5
{
    public static void main(String[] args)
    {
        Singleton single1 = Singleton.getInstance();
        Singleton single2 = Singleton.getInstance();

        System.out.println(Singleton.count);
        System.out.println(single2.count);
        single1.show();

    }

}
```

Topic 06 [**Java Multithreading**]

Problem Statement: Write a program to run three applications i) Word Processor ii) Music Player and iii) Real-time clock simultaneously. Use run() method of Thread class or implements runnable interface.

```
class Wordprocessor extends Thread
{
    public void run()
    {
        for(int i=0; i<10; i++)
        {
            System.out.println("Word processor is running");
        }
    }
}
class Musicplayer extends Thread
{
    public void run()
    {
        for(int i=0; i<10; i++)
```

```
        {
            System.out.println("Musicplayer is running");
        }
    }
}
class Realtimeclock extends Thread
{
    public void run()
    {
        for(int i=0; i<10; i++)
        {
            System.out.println("Timer is on");
        }
    }
}
public class Topic6 {
    public static void main(String[] args)
    {
        Wordprocessor word = new Wordprocessor();
        Musicplayer music = new Musicplayer();
        Realtimeclock timer = new Realtimeclock();
        word.start();
        music.start();
        timer.start();

    }

}


class words implements Runnable
{
    public void run()
    {
        for(int i=0; i<10; i++)
        System.out.println("Ms Word is running");
    }
}
class music implements Runnable
{
    public void run()
    {
        for(int i=0; i<10; i++)
        System.out.println("Music is playing");
    }


}
class timer implements Runnable
```

```
{
    public void run()
    {
        for(int i=0; i<10; i++)
        System.out.println("Time is ticking");
    }
}
public class Topic6runnable
{
    public static void main(String[] args)
    {
    words w = new words();
    Thread w1= new Thread(w);
    music m = new music();
    Thread m1 = new Thread(m);
    timer t = new timer();
    Thread t1 = new Thread(t);
    w1.start();
    m1.start();
    t1.start();
    }

}
```

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering
24 Series

Course No.: CSE 1203
Course Title: Object Oriented Programmming

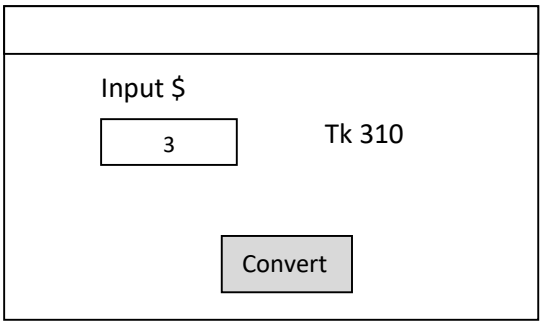
Experiment No. : 09
Experiment Title : Advanced Topic: Java Swing Programming

Submitted by:	**Submitted to:**
Muhammad Ragib Ishrak Roll: 2403013 Section: A	Shyla Afroge Associate Professor Dept. of Computer Science & Engineering, RUET

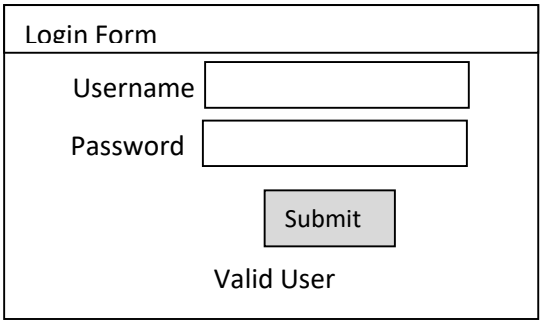
Date of Experiment :
Date of Submission : 26 April 2026

Module 9
[Advanced Topics: Java Swing Programming] [CO4]

Topic 1[**java swing: Money Changer**]
Problem Statement: Write a Money Changer (Dollar to Taka) program in Java with the following components. When user clicks **Convert** button, its shows the output in a label.



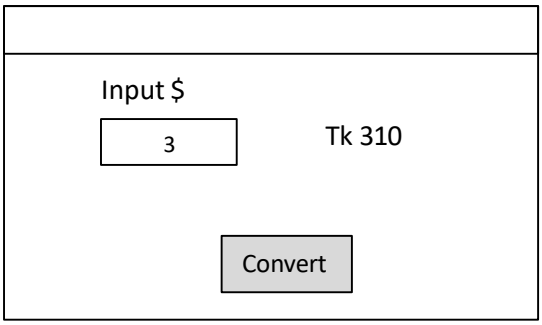
Topic 2[**java swing: Login Form**]
Problem Statement: Write a Java program that accepts username and password of user and then displays whethere the user is valid or NOT., It shows the output in a label.



Module 9
[Advanced Topics: Java Swing Programming]

Topic 1[java swing: Money Changer]

Problem Statement: Write a Money Changer (Dollar to Taka) program in Java with the following components. When user clicks **Convert** button, its shows the output in a label.

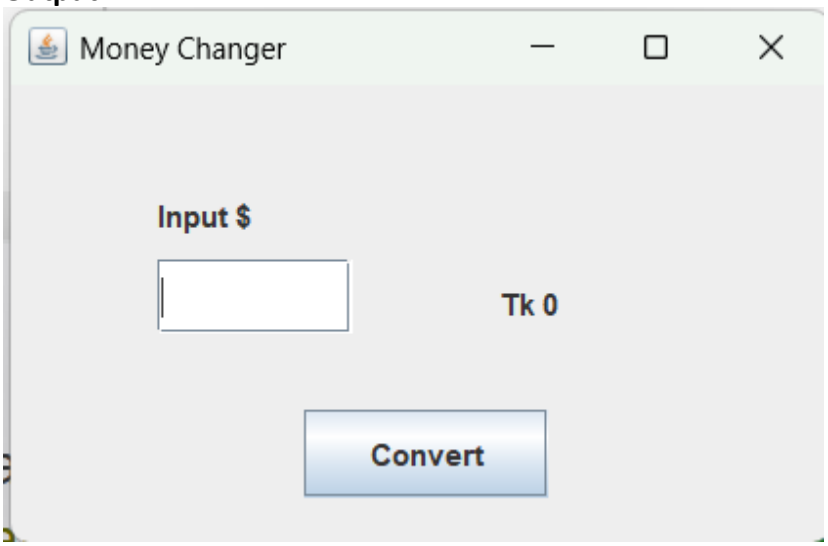


Code:

```
public class MoneyChanger {
    public static void main(String[] args) {
        javax.swing.JFrame frame = new javax.swing.JFrame("Money Changer");
        frame.setSize(350, 220);
        frame.setDefaultCloseOperation(javax.swing.JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);

        javax.swing.JLabel inputLabel = new javax.swing.JLabel("Input $");
        inputLabel.setBounds(60, 40, 80, 25);
        frame.add(inputLabel);

        javax.swing.JTextField dollarField = new javax.swing.JTextField();
        dollarField.setBounds(60, 70, 80, 30);
        frame.add(dollarField);

        javax.swing.JLabel resultLabel = new javax.swing.JLabel("Tk 0");
        resultLabel.setBounds(200, 75, 100, 25);
        frame.add(resultLabel);

        javax.swing.JButton convertButton = new javax.swing.JButton("Convert");
        convertButton.setBounds(120, 130, 100, 35);
        frame.add(convertButton);

        convertButton.addActionListener(e -> {
            try {
                double dollar = Double.parseDouble(dollarField.getText());
                double taka = dollar * 110;   // 1 Dollar = 110 Taka
                resultLabel.setText("Tk " + (int)taka);
            } catch (NumberFormatException ex) {
```

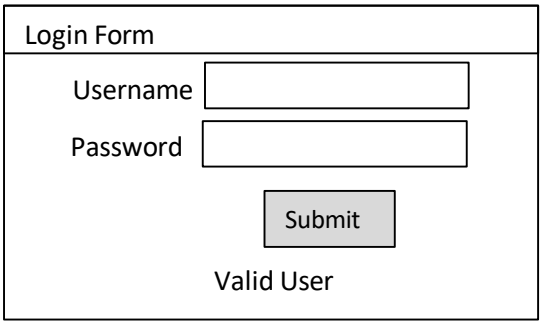
```
                resultLabel.setText("Invalid Input");
            }
        });

        frame.setLocationRelativeTo(null);
        frame.setVisible(true);
    }
}
```

Output:



Topic 2[**java swing: Login Form**]

Problem Statement: Write a Java program that accepts username and password of user and then displays whethere the user is valid or NOT., It shows the output in a label.

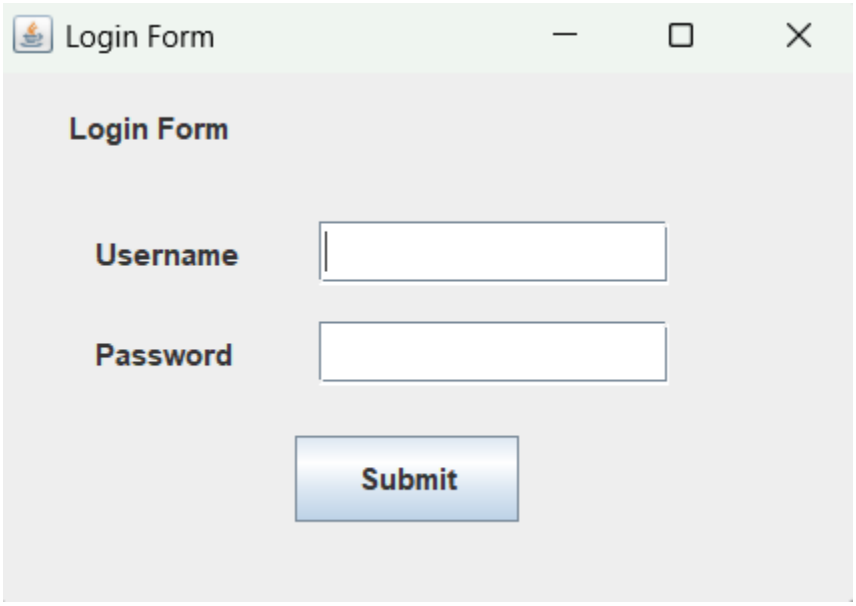


Code:

```
public class LoginForm {
    public static void main(String[] args) {
        javax.swing.JFrame frame = new javax.swing.JFrame("Login Form");
        frame.setSize(360, 250);
        frame.setDefaultCloseOperation(javax.swing.JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);

        javax.swing.JLabel titleLabel = new javax.swing.JLabel("Login Form");
        titleLabel.setBounds(30, 10, 100, 25);
```

```
        frame.add(titleLabel);

        javax.swing.JLabel userLabel = new javax.swing.JLabel("Username");
        userLabel.setBounds(40, 60, 80, 25);
        frame.add(userLabel);

        javax.swing.JTextField userField = new javax.swing.JTextField();
        userField.setBounds(130, 60, 140, 25);
        frame.add(userField);

        javax.swing.JLabel passLabel = new javax.swing.JLabel("Password");
        passLabel.setBounds(40, 100, 80, 25);
        frame.add(passLabel);

        javax.swing.JPasswordField passField = new javax.swing.JPasswordField();
        passField.setBounds(130, 100, 140, 25);
        frame.add(passField);

        javax.swing.JButton submitButton = new javax.swing.JButton("Submit");
        submitButton.setBounds(120, 145, 90, 35);
        frame.add(submitButton);

        javax.swing.JLabel resultLabel = new javax.swing.JLabel("");
        resultLabel.setBounds(120, 185, 120, 25);
        frame.add(resultLabel);

        submitButton.addActionListener(e -> {
            String username = userField.getText();
            String password = new String(passField.getPassword());

            if (username.equals("admin") && password.equals("1234")) {
                resultLabel.setText("Valid User");
            } else {
                resultLabel.setText("Invalid User");
            }
        });

        frame.setLocationRelativeTo(null);
        frame.setVisible(true);
    }
}
```

Output:


Login Form
Login Form
Username
Password
Submit